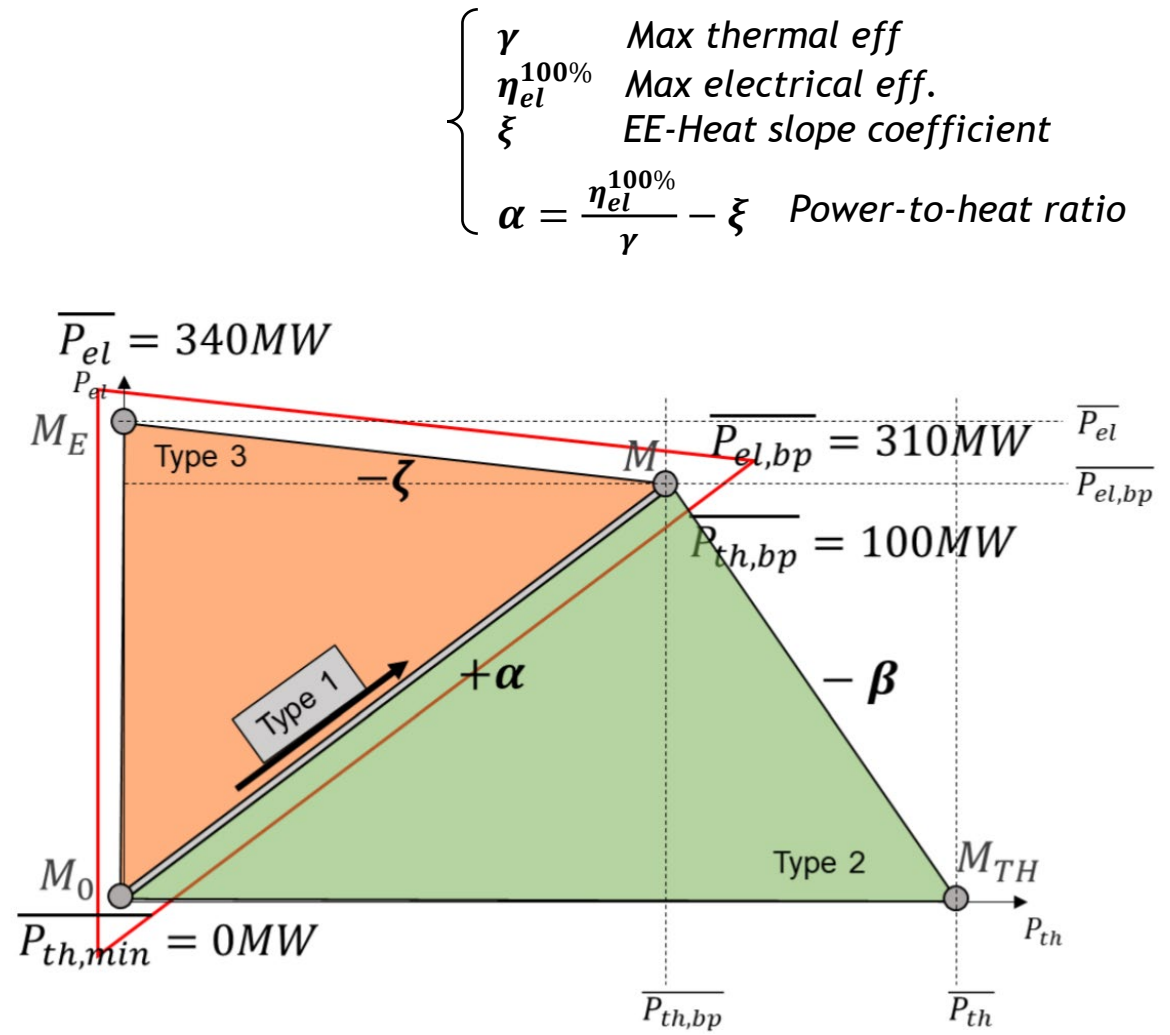
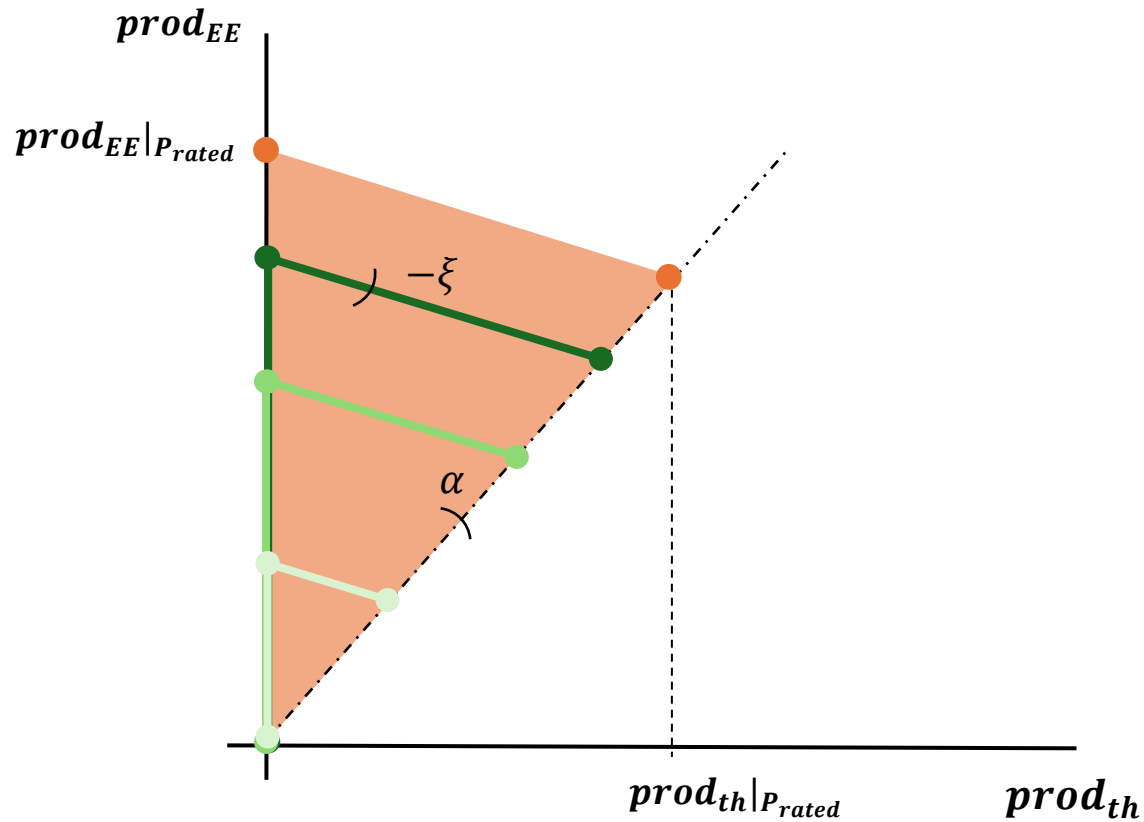


Cogeneration Modes: Type 3



$$\begin{cases} \gamma & \text{Max thermal eff} \\ \eta_{el}^{100\%} & \text{Max electrical eff.} \\ \xi & \text{EE-Heat slope coefficient} \\ \alpha = \frac{\eta_{el}^{100\%}}{\gamma} - \xi & \text{Power-to-heat ratio} \end{cases}$$

Equations ruling Variable Range Cogeneration in Hypatia

- $prod_{EE} = \eta_{el}^{100\%} \cdot prod_{CHP} - \xi \cdot prod_{th}$ $= prod_{EE}|_{P_{rated}}$
- $prod_{th} \leq \gamma \cdot prod_{CHP}$ $= prod_{th}|_{P_{rated}}$
- $\alpha \cdot prod_{th} \leq prod_{EE}$



How to insert parameters to make CHP cluster work

e.g. RES with variable range CHP techs

